COURSE PLAN

FIRST: BASIC INFORMATION

College

College	: Prince Abdullah bin Ghazi Faculty of ICT
Department	: Computer Science

Course

Course Title	: Compiler Design and programming languages
Course Code	: 30801417
Credit Hours	: 3
Prerequisite	: 30801316

Instructor

Name	: Dr. Hasan Rashaideh
Office No.	: 2nd Floor Room 502
Tel (Ext)	: -
E-mail	: rashaideh@gmail.com
Office Hours	: Su-Mo-Tu- (11:00-12:00AM)

Class Times

Building	Day	Start Time	End Time	Room No.
IT	Su-Mo-Tu-We	09:00	10:00am	Lab302

Text Book

Title	Compilers Principles, Techniques, and Tools. 2nd Edition, Aho, Sethi, and Ullman, Addison Wesley, 2006.

References

1- Modern Compiler Implementation in Java by A.W. Appel. Cambridge University Press, 2002.
2- Modern Compiler Design by D. Grune, H.E. Bal, C.J.H. Jacobs, K.G. Langendoen. Wiley, 2012.

SECOND: PROFESSIONAL INFORMATION

COURSE DESCRIPTION

Design and structure of complex grammars, lexical analysis, parsers, semantic data structures and code generation and optimization. Construction of a simple compiler.

COURSE OBJECTIVES

1. Gain an understanding of how compilers translate source code to machine executable.
2. Utilize tools to automate compiler construction
3. Comprehend how to perform parsing (top down and bottom up).
4. Have the knowledge to design, implement, and test a compiler for a simple language

COURSE LEARNING OUTCOMES

Knowledge and Understanding

- Acquire a full understanding and mentality of Automata Theory as the basis of all computer science languages design
- Have a clear understanding of the Automata theory concepts such as RE's, DFA's, NFA's, and Grammars.
- Design FAs, NFAs, Grammars, languages modelling, small compilers basic
- Minimize FA's and Grammars of Context Free Languages
- Ability to create lexical rules and grammars for a programming language.
- Ability to use Flex or similar tools to create a lexical analyzer and Yacc/Bison tools to create a parser.
- Ability to implement a lexer without using Flex or any other lexer generation tools.
- Ability to implement a parser such as a bottom-up SLR parser without using Yacc/Bison or any other compiler-generation tools.

Professional Skills

- to recognize the significance of complexity classes and analysis and to deduce the complexity of certain types of algorithm
- to demonstrate a deeper and broader understanding of classes of complexity
- to prove the basic results of the Theory of Computation

Competences (Transferable skill and attributes)

- Search for, analysis and synthesis of data and information, with the use of the necessary technology

COURSE SYLLABUS

Week	Course Topic	Notes
Week 1	**Introduction, Language Processors, and The Structure of a Compiler.**	**Ch1 and Ch2**
Week 2	**Review of Formal Languages, Regular expressions, NFA, DFA**	**Ch3**
Week 3	**RE to NFA, NFA to DFA, and DFA to RE**	**Ch3**
Week 4	**Minimizing DFA, Lexical Analysis**	**Ch3**
Week 5	**JFLEX tools**	**Ch3**
Week 6	**Parsing, Context Free Grammars, NFA to CFG , derivations.**	**Ch4**
Week 7	**Ambiguity, Top Down (LL) Parsing, Bottom up (LR) Parsing, FIRST, FOLLOW, and PREDICT sets**	**Ch4**
Week 8		**Midterm Exam**
Week 9	**LL Parsing: Recursive Descent parsing, and predictive parsing**	**Ch4**
Week 10	**LL(1) Grammar: common prefixes, left recursion**	**Ch4**
Week 11	**LR Parsing: Shift Reduce Parsing, Action and GoTo Tables,**	**Ch4**
Week 12	**Generating SLR parse tables**	**Ch4**
Week 13	**LR Conflicts, Error Recovery in LR parsing**	**Ch4**
Week 14	**LR(1) parsing**	**Ch4**
Week 15	**LR(1) grammar , LALR approach**	**Ch4**
Week 16	**Final Exam**	**Final Exam**

COURSE LEARNING RESOURCES

Teaching materials for students (handouts, exercise collections, examples of solved exercises), detailed course syllabus, and all other data are available by e-learning MOODLE system to the registered students via elearning.bau.edu.jo.

ONLINE RESOURCES

- NA

ASSESSMANT TOOLS

(Write assessment tools that will be used to test students ability to understand the course material and gain the skills and competencies stated in learning outcomes

ASSESSMENT TOOLS	%
Quizzes	5
Assignments	5
Mid Exam	40
Final Exam	50
TOTAL MARKS	100

COURSE COORDINATOR

Course Coordinator: Dr. Hasan Rashaideh	**Department Head: Dr. Omar Alzubi**
Signature:	**Signature:**
Date:	**Date:**